

Standard Operating Procedure

Ladder Safety – Commercial & Residential Installations

Document No.	SOP-LV-009
Effective Date	July 13, 2026
Revision	1.0
Review Date	July 13, 2027
Applicable Sites	All commercial, institutional, and residential job sites (Ontario & Manitoba)
Prepared By	_____ (Operations / H&S Lead)
Approved By	_____ (Owner / Principal)

1. Purpose

This procedure establishes minimum safe practices for the selection, inspection, setup, and use of portable ladders (step and extension) by technicians performing low-voltage and structured cabling work at commercial, institutional, and residential sites. It is intended to reduce the risk of falls, electrical contact, and struck-by incidents during ladder use.

2. Scope

Applies to all employees, subcontractors, and site supervisors who use portable ladders while performing cable installation, termination, device mounting, or related low-voltage work. Covers step ladders, single/extension ladders, and combination ladders. Does not cover scaffolding, aerial lifts, or fixed/permanent ladders — see separate procedures where applicable.

3. Definitions

- **Duty Rating:** The maximum safe working load for a ladder, including the worker, tools, and materials, as specified by the manufacturer (e.g., CSA Grade 1, 1A, 1AA).
- **4-to-1 Rule:** For every 4 feet of working height, the base of an extension/straight ladder must be set out 1 foot from the vertical support surface.
- **Three Points of Contact:** Maintaining two hands and one foot, or two feet and one hand, in contact with the ladder at all times while climbing.
- **Competent Worker:** A worker who is qualified by training and experience to recognize ladder hazards and has knowledge of applicable regulations.

4. Responsibilities

Site Supervisor / Lead Technician

- Confirms only inspected, tagged ladders are brought on site.
- Assesses ground conditions, overhead hazards, and traffic before ladder placement.
- Stops work if unsafe ladder conditions are identified.

Technician

- Performs a pre-use inspection before every use, each shift.
- Sets up and uses the ladder in accordance with this SOP and manufacturer instructions.
- Reports damaged, unstable, or missing ladder equipment immediately and removes it from service.

5. Pre-Use Inspection

Before each use, inspect the ladder for:

- Cracked, bent, corroded, or splintered rails and rungs.
- Loose, missing, or damaged rung locks, spreader braces, or hinges.
- Worn, missing, or slipping non-slip feet.
- Frayed or damaged rope on extension ladders.

- Legible duty rating and CSA/manufacturer label.
- Excessive dirt, grease, or paint that could obscure defects or affect grip.

Any ladder failing inspection is tagged “Do Not Use,” removed from service, and reported to the site supervisor immediately.

STOP WORK:

Never use a wood or fiberglass-only rule as a substitute for a damaged ladder. Metal or conductive ladders are prohibited within reach of energized electrical equipment, overhead lines, or panels — use fiberglass ladders for all electrical proximity work.

6. Ladder Selection & Setup

Selection

- Select a ladder with a duty rating that exceeds the combined weight of the worker, tools, and materials.
- Use fiberglass (non-conductive) ladders for any work near electrical panels, energized conduit, or overhead services.
- Choose a ladder length that allows the worker to stand no higher than the second rung from the top (step ladder) or 3 feet below the top (extension ladder).

Setup

- Inspect the surface — firm, level, and free of debris, oil, or standing water.
- Apply the 4-to-1 rule for straight/extension ladders.
- Fully open and lock spreader braces on step ladders before climbing.
- Extend an extension ladder at least 3 feet (1 m) above the landing surface when used for access to a roof or platform.
- Secure the ladder at the top, bottom, or both when working conditions or duration require it (tie-off, stakes, or a second worker footing the ladder).
- Never set up a ladder on boxes, tables, scaffolding, or other unstable surfaces to gain additional height.

7. Commercial & Institutional Site Considerations

Applies to schools, offices, and other occupied or public-facing sites.

- Coordinate ladder placement with the site contact to avoid occupied classrooms, hallways during transition periods, and public traffic areas.
- Barricade or cone off the ladder work area and post warning signage when working in corridors, lobbies, or other through-traffic zones.
- Never leave an unattended ladder set up in an area accessible to students, staff, or the public — lower or remove it during breaks.
- Confirm overhead clearances against suspended ceiling grids, sprinkler heads, fire/smoke detectors, and lighting before raising extension ladders.
- Where a ladder must be placed near a doorway, lock or barricade the door and post signage, or assign a spotter.

8. Residential Site Considerations

Applies to occupied private residences.

- Confirm with the homeowner before ladder placement near landscaping, driveways, decks, or uneven yard grading.
- Be aware of pets and children who may approach or disturb a set-up ladder; ask the homeowner to secure pets during the visit.
- Use extra caution on interior stairwells — use a stairway-rated ladder or leg-leveling ladder designed for uneven stair surfaces; never set a standard ladder on stair treads.
- Inspect exterior ground conditions for soft soil, slopes, or hidden hazards (sprinkler heads, window wells) before setup.
- Do not rely on homeowner-supplied ladders; use only company-inspected equipment.

9. Climbing & Working Practices

- Maintain three points of contact at all times while climbing and descending.
- Face the ladder when climbing or descending; never climb with your back to the ladder.
- Use a tool belt, bucket hook, or hoist line to raise/lower tools and materials — do not carry items in hand while climbing.
- Keep your belt buckle (navel) between the ladder side rails; do not overreach. Reposition the ladder instead of leaning.
- Only one worker on a ladder at a time.

- Do not stand on the top step or top cap of a step ladder, or the top three rungs of an extension ladder.
- Do not use a ladder in high wind, near doorways that could open into it, or on wet/icy surfaces without corrective measures.

10. Electrical & Overhead Hazard Awareness

- Maintain minimum approach distances from energized overhead power lines per applicable electrical safety authority requirements; treat all overhead lines as energized.
- Use non-conductive (fibreglass) ladders whenever work is performed near electrical panels, energized equipment, or overhead services.
- De-energize and lock out circuits where feasible before working in close proximity with a ladder.
- Be alert to low-voltage and line-voltage conduit, junction boxes, and fixtures concealed above drop ceilings before reaching into ceiling spaces from a ladder.

11. Fall Protection

Ladders are not a substitute for fall protection where working height and duration warrant additional controls.

- Where sustained work above 3 m (10 ft) is required, evaluate the use of a mobile platform, scaffold, or aerial lift instead of a ladder.
- Follow the employer's fall protection plan and applicable occupational health and safety regulation requirements for the jurisdiction and work height involved.
- Do not use a ladder as a work platform for extended, repetitive-motion tasks — reposition frequently rather than overreaching.

12. Prohibited Practices

- Using a metal ladder near any electrical hazard.
- Using a ladder with a rating below the combined worker/tool/material weight.
- Standing on the top step, top cap, or paint shelf of a step ladder.
- Splicing or joining two ladders to gain additional height.
- Using a damaged or "Do Not Use" tagged ladder for any reason.
- Leaving a set-up ladder unattended in an occupied or public area.

13. Required PPE

- CSA-approved footwear with slip-resistant soles.
- Hard hat where overhead hazards are present (per site H&S checklist, SOP-LV-003).
- Safety glasses when working overhead or drilling/cutting above head height.
- Fall protection harness and lanyard where required by height, duration, or site policy.

14. Emergency Procedures

- In the event of a fall or ladder-related injury, stop work, secure the area, and provide first aid/call 911 as needed.
- Report all incidents to the site supervisor immediately and complete an incident report consistent with company procedure.
- Preserve the ladder and setup conditions where safe to do so pending review.

15. References

- Ontario Occupational Health and Safety Act (OHSA) and O. Reg. 213/91 (Construction Projects).
- Ontario O. Reg. 851 (Industrial Establishments), where applicable to institutional sites.
- CSA Standard Z11 – Portable Ladders.
- Manitoba Workplace Safety and Health Act and Workplace Safety and Health Regulation (ladders and fall protection provisions), for Manitoba job sites.
- WSIB / applicable workers' compensation reporting requirements.

Note: Regulatory citations are provided for reference only; confirm current section numbers and requirements against the applicable jurisdiction's up-to-date regulations before relying on this document for compliance purposes.

16. Technician Sign-Off

By signing below, I confirm that I have read, understood, and will comply with this Standard Operating Procedure (SOP-LV-009 – Ladder Safety – Commercial & Residential Installations).

Printed Name	Signature	Date	Site / Project

Revision History

Rev. 1.0 — 2026-07-13 — Initial release.